SESSION 01

Bank Branch Security Lab

Phase 1 · Network perimeter · October 7, 2026

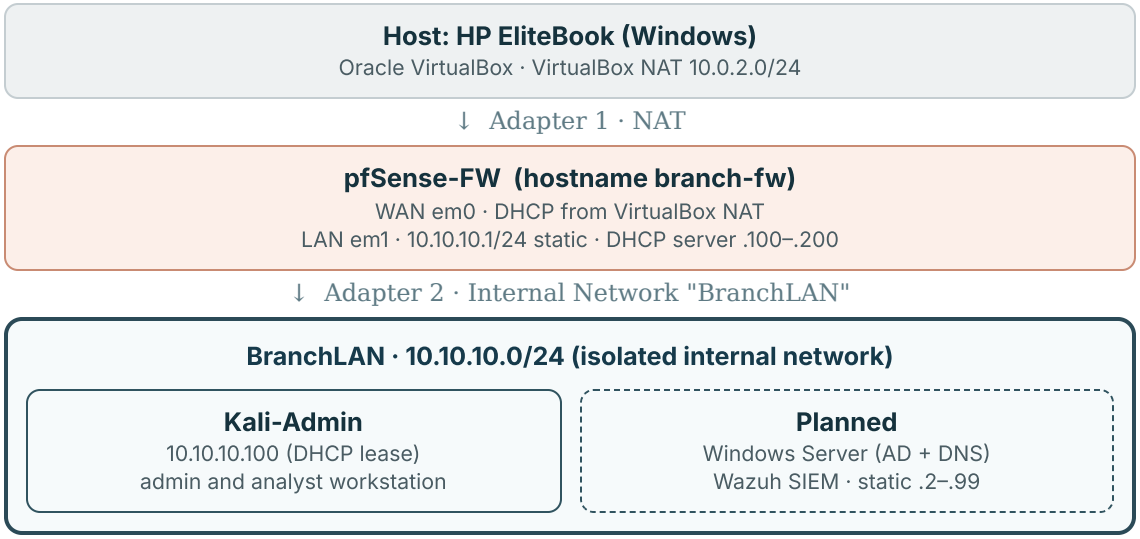
Goal	Stand up the network perimeter for a simulated bank branch: a firewall that separates the branch network from the internet, plus an admin workstation inside it.
Outcome	Firewall deployed and hardened, branch LAN segmented and serving DHCP, Kali admin workstation connected and verified.
Systems built	pfSense-FW (pfSense CE 2.9.0) · Kali-Admin (Kali Linux 2026.2)
Branch network	10.10.10.0/24 · gateway 10.10.10.1 · DHCP pool 10.10.10.100–200
Skills	Virtualization, firewall deployment, IP addressing and subnetting, DHCP, network segmentation, baseline hardening

Overview

This phase established the foundation of the lab. On Oracle VirtualBox I deployed a pfSense firewall with two interfaces: a WAN that reaches the internet through VirtualBox NAT, and a LAN on an isolated internal network named BranchLAN (10.10.10.0/24, gateway 10.10.10.1, with a DHCP pool). A Kali Linux workstation was attached only to BranchLAN, which forces all of its traffic through the firewall. I completed the initial setup wizard, replaced the default administrator password, configured time synchronization, and enabled spoofed-address (RFC 1918 and bogon) blocking on the WAN.

Architecture

A standard perimeter model: one firewall with an untrusted side (WAN) and a trusted side (LAN). Every packet between the branch and the internet must cross the firewall, where it can be allowed, blocked, and logged.



IP addressing plan

Item	Value	Notes
WAN (em0)	DHCP from VirtualBox NAT	NAT network 10.0.2.0/24; pfSense receives 10.0.2.15
LAN (em1)	10.10.10.1 / 24 (static)	Default gateway for every branch device
Branch subnet	10.10.10.0/24	256 addresses, 254 usable hosts
Reserved for servers	10.10.10.2 – 10.10.10.99	Static addresses for infrastructure
DHCP pool	10.10.10.100 – 10.10.10.200	Handed out to workstations
Admin access	https://10.10.10.1	HTTPS web console, reachable from the LAN only

What I built

1. Chose Oracle VirtualBox as the hypervisor after the VMware Workstation download failed; it is free and sufficient for a multi-VM lab.
2. Provisioned the pfSense firewall VM (FreeBSD-based, 2 GB RAM, 1 vCPU, 20 GB disk) with dual-homed networking: Adapter 1 on NAT (WAN), Adapter 2 on the internal network BranchLAN (LAN).
3. Installed pfSense CE 2.9.0; the installer downloaded and installed 179 packages, which also confirmed WAN connectivity.
4. Diagnosed and fixed a post-install boot loop by detaching the installer ISO and correcting the VM boot order.
5. Addressed the LAN as 10.10.10.1/24 and enabled the DHCP server with a 10.10.10.100–200 pool, keeping the web console on HTTPS.
6. Imported the Kali workstation and moved its only adapter to BranchLAN, so it reaches the internet only through the firewall.
7. Verified addressing: Kali received 10.10.10.100 and reached the pfSense console at https://10.10.10.1.
8. Completed the setup wizard: hostname branch-fw, NTP and time zone, WAN on DHCP with RFC 1918 and bogon blocking enabled, and the default admin password replaced.

Design decisions

Decision	Rationale
Build the firewall first	Every other system sits behind it. Infrastructure others depend on is built first.
Two adapters on the firewall	Creates two zones (untrusted WAN, trusted LAN) with the firewall as the only path between them: a single controlled choke point.
10.10.10.0/24, not 192.168.1.0/24	Most home routers use 192.168.1.x. A distinct private range avoids overlapping-subnet routing conflicts.
Static gateway, DHCP for clients	Clients need a fixed, known exit point. Workstations are addressed automatically, which scales and avoids typos.
DHCP pool starts at .100	Leaves .2–.99 free for servers that need permanent addresses, such as the domain controller.
Kali on BranchLAN only (no NAT)	Mirrors a real branch PC, which cannot bypass the firewall. All traffic is filtered and later logged.
HTTPS for the web console	Encrypts administrator credentials in transit; HTTP would send them in clear text.

Verification

Check	How	Result
Firewall reaches the internet (WAN)	Installer downloaded 179 packages	Pass
Firewall boots from its own disk	Console menu appears after reboot	Pass
Branch DHCP works	`ip a` on Kali shows 10.10.10.100/24	Pass
Web console reachable from LAN only	https://10.10.10.1 from Kali	Pass
Default admin password replaced	Setup wizard; warning banner gone	Pass
Kali reaches the internet through the firewall	`ping -c 4 8.8.8.8`	Pass
DNS resolution works	`ping -c 4 google.com`	Pass

Troubleshooting

Firewall boot loop after installation

Cause	The installer ISO was still attached and the optical drive came before the hard disk in the boot order, so the VM kept booting the installer.
Fix	Removed the ISO from the virtual optical drive and moved the hard disk to the top of the boot order. Fix the cause, not just the symptom.

Unexpected root shell at the end of installation

Cause	Pressing **S** triggered the dialog's keyboard shortcut for Shell.
Fix	Typed `exit` to return to the dialog. In text-mode installers, navigate with arrow keys, Tab and Enter only.